

Detecting Users with Multiple Aliases on Twitter

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Abstract—In day to day life, many people create their multiple identities on social networking sites in order to abstract their true identity. No wonder, monitoring and analysis of the social networks are becoming important for analysts since terrorists and extremists are using the information provided by the user, to spread advocacy and communicating with each other. As Twitter is a widely used social network to share opinions and views, in this work, we have applied Supervised Machine Learning (i.e. alias detection) on the Twitter dataset. Here, we propose a technique for detecting users who make use of aliases in which time profiles are incorporated with the stylometric-based features. By combining time profiles and stylometric features, the detection rate of a user having multiple aliases increases significantly. Further, we make use of data mining techniques to improve accuracy. The obtained results show that time profiles can pose to be very useful tools for alias detection when combined with stylometry.

Keywords: Social network; Alias detection; Machine learning; Data mining; Time profiles; Stylometric-based features.

I. INTRODUCTION

Social media usage has increased rapidly over the years. The social networking sites in the world of today are able to attract people from different parts of the world to interact with each other and share their opinions, ideas, personal judgments on any subject matter [8]. Usually, the information is expressed publicly and is intended for public reading.

But many extremists groups are taking advantage of this facility for spreading sensitive views and information [1]. The usage of printed magazines and centralized websites have been largely replaced by many interactive discussion forums and social media platforms. Common discussion forums used by extremists include Ansar Al-Mujahidin Jihadist Forum, Twitter etc. along with other social media platforms, e.g., YouTube, Facebook [2][3].

It is possible to monitor the activities of these terrorists by monitoring the traffic to forums and websites. Users accessing these websites can be identified with the help of their unique IP (Internet Protocol) addresses. As most of the individuals behind these extremist activities do not have a unique identification and usually, have multiple aliases on the social networks along with an ease to be able to frequently alter the location of their Web Hosting server, this imposes difficulty for monitoring their behavior.

Identifying the aliases of the terrorist groups or the extremists can help to analyze their behavior and provide additional support towards cyber security. Since an individual can not only have a specific pattern of writing but can also have

a pattern of posting content on the discussion forums, this time-specific information can be vital in the identification of the aliases. Aliases are also formed by the ordinary citizens who try to preserve their anonymity in certain situations such as when the discussion on a web forum is on critical topics, e.g., sexual preferences, political ideas, religion. Consideration of the information as private or sensitive is bound to vary from individual to individual and country to country. In that scenario too, alias identification can be extended.

Timeprint is a property that can explain the behavioral pattern of an individual. Various time features also help to identify the identity of an individual online. Timeprint can be related to the publishing times of the posts by a user. Using timeprints, message distribution can be examined over a period of days, hours of a day, weekdays so on. The timeprints along with stylometry can show significant increase in the detection of the multiple aliases. The timeprints, in association with the stylometry, in many cases, can develop more powerful feature set.

II. LITERATURE REVIEW

Here, few literature attempted to overcome the difficulties related to online social networks where users can create several aliases and cover up their identity. In [6], first implementation of time profiles on the forum dataset is performed to analyze the impact of time profiles in identifying the users having multiple aliases. Hence, timeprints is proved the as a powerful tool. Narayanan et. al [9] propose a method for identifying anonymous user via stylometric based matching and compare the known authorship of users based on their writing styles. Dahlin et. al [4] perform analysis of writing style of users using stylometric features which are topic independent. In this work, top correct authors are identified with much good accuracy. Kourtis et. al [7] introduce two major categories, profile based models and instance based models and purpose an approach to combines well-known representative of these two categories n-Grams method and SVM classifier. The existing earlier work is limited to the Irish discussion forum. No work has been done on Twitter, which is a widespread social networking site used all over the world, allows its users to share their opinions and views. The earlier works done by various authors do not consider time prints along with the stylometric features for Twitter to identify the multiple aliases.

In this work, we develop techniques for identification of Twitter users having multiple aliases and analyze the

importance of time profiles along with stylometric features in order to improve the accuracy. In addition, we examine the classification quality of SVM classifier over Naive Bayes for the combination of time profiles and stylometry. Rest of the paper is organized as follows. Section 2 presents proposed method. Section 3 presents a comparative study and discussion of the results. Finally, section 4 concludes and remarks upon the possible future research directions.

III. PROPOSED METHOD

A. Stylometric Technique for Alias Detection

Our stylometric implementation includes a subset of features used in [2]. The features which have been used are summarized in Table I.

Along with the features of this subset, other features, e.g., vocabulary richness of Hapax Legomena or Yules K and syntactic features, could also have been used. We have rather incorporated a rich set of possible features that are reasonably fast and can be extracted from the Twitter posts. Since to get data with multiple aliases is hard, the extracted dataset of Twitter is used to synthetically create the data. The users with large number of posts are hence thereby used as this helps to increase the data material as much as possible. Since to get data with multiple aliases is hard, the extracted dataset of Twitter is used to synthetically create the data. The users with large number of posts are hence thereby used as this helps to increase the data material as much as possible.

- In this experiment, we have considered top-50 users who have more than 100 posts.
- A user U_i has been split into five sub-users U_{i1} , U_{i2} , U_{i3} , U_{i4} and U_{i5} .
- The posts are then sequentially distributed among the sub-users that is, the first post is assigned to U_{i1} and second to U_{i2} and so on. Now, user U_i has four other aliases. These five sub-users facilitate construction of several training instances for each user. This helps in the learning phase of supervised learning. The user ID, U_i is then set as the target class. Each class has five different data instances. Supervised Learning algorithms used for computing the accuracy of the experiment:
 - Naive Bayes classifier
 - Support Vector Machine classifier

TABLE I. THE STYLOMETRIC FEATURES

Category	Description	Count
Letters	Relative Frequency of a-z(ignoring case)	26
Digits	Relative Frequency of 0-9	10
Special Characters	Relative Frequency of characters . , : ? etc.	18
Words	Relative Frequency of words with 1-20 characters	20

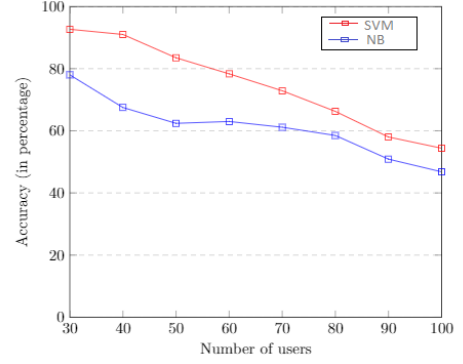


Fig. 1. Accuracy Results for stylometric technique from classification with NB and SVM classifiers

B. Combined Technique for Alias Detection

In this part of the experiment, along with the features used in the previous experiment, additional time features are also added for detection of aliases. The time profile of a user is constructed by using the following set of features when constructing our time prints:

- Hour of the day (hour1, hour2, ..., hour24)
- Period of day (MidNight, EarlyMorning, Morning, MidDay, Evening, Night)
- Month (January, February, ..., December)
- Day (Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday)
- Type of Day (Weekday, Weekend).

Similar to the previous experiment, supervised learning is performed. For the top-50 users, five sub-users are created sequentially which corresponds to the five instances of the target class U_i . The Naive Bayes and SVM classifier are used, trained and the corresponding results are computed. The inclusion of the time profiles of the users shows better results than the individual stylometric technique. The previous two matching techniques are not only used for top-50 users. Incrementally, the number of users are increased by 10 to 60 and so on till 100. Then, this is also applied to top-40, top-30 users respectively for analysis of the result pattern. The following procedure is carried out in order to validate our research, also as shown in Fig 1:

We arrange the dataset in the descending order of number of posts. The top 50 users are picked from dataset having larger number of posts (i.e. more than 100). Each user is split into five sub-users. Assume that we have taken User1 from the dataset and then split it into U_{i1} , U_{i2} , U_{i3} , U_{i4} and U_{i5} . The posts and time print of User1 are also divided between these five sub-users sequentially.

After this, for stylometric alias detection, we compute the frequency vector for each of the four features namely alphabets, digits, special characters, word length, and then categorize these vectors on the basis of users into a single database. Then we apply classifiers on this dataset to get the accuracy of detecting potential aliases. In case of time prints, we do the same for the time print features: period of day, hour, type of

day, week, month, date.

For the combination of stylometric-based and time prints alias detection, we combine the database obtained from stylometry and time prints individually and then get the accuracy by applying classifiers (i.e. Naive Bayes and SVM). The above test procedure is repeated for the top 30 users and top 40 users respectively. Similarly, the users are incremented from 50 to 100 in step of 10 users each at a time, picking the users with highest number of posts.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Data Set

The Twitter dataset of our proceeding work was gathered from the Twitter sentiment corpus of Sentiment140 as mentioned in [5]. This dataset contains total 1600000 entries of tweets for the users. We made algorithms to pre-process this dataset for the purpose of training and testing. As Microsoft Excel was able to open only 500000 of data, we ran a python script to extract complete data through Python.

Our dataset of Twitter includes the tweets of various users from May to June, 2009. Format of the original data file (6 fields):

- The polarity of the tweet
- The id of the tweet
- The date of the tweet
- The query. For no query, this value is NO QUERY
- The user that tweeted
- The text of the tweet.

This dataset is further pre-processed using Python and the number of posts for all the users are calculated for each user. Then these users are listed in the decreasing order of the number of the posts as per the experimental requirements.

B. Training and Testing the classifier

The experimental analysis of the proposed work has been done by dividing the dataset into training-testing sets in the ratio of (80-20) and (70-30). We basically used two classifiers for this work: SVM and Naive Bayes classifier. 10-fold cross validation technique is used for validating the training phase of the classifier. All the work of training and testing classifier has been done on WEKA. We have conducted different experiments on the dataset to understand the influence of time prints on the author identification and alias matching.

C. Results Analysis

The experiment shows the identification of the aliases of a Twitter user using supervised learning techniques. The stylometric analysis shows that for the top30 users, the Naive Bayes classifier gives an accuracy of 78% and the Support Vector Machine classifier gives an accuracy of 92.66% as depicted in the Table II.

TABLE II. ACCURACY RESULTS FOR STYLOMETRIC TECHNIQUE

Users	Stylometry (NB)	Stylometry (SVM)
30	78%	92.67%
40	67.50%	91%
50	62.40%	83.50%
60	63%	78.33%
70	61.14%	72.88%
80	58.50%	66.25%
90	50.88%	58%
100	46.80%	54.40%

TABLE III. ACCURACY RESULTS FOR COMBINED TECHNIQUE

Users	Combined (NB)	Combined (SVM)
30	76%	92.50%
40	69.50%	92.50%
50	66.80%	88.40%
60	61.33%	87.66%
70	65.66%	83.71%
80	59%	81%
90	55.33%	68.70%
100	50.60%	67%

As the number of users are increased from 30 to 40, there is a decrease in the accuracy to 67.5% in case of the Naive Bayes classifier and 91% in the case of the Support Vector Machine classifier. On further increase in the number of users to 50, the accuracy of correctly identifying the aliases based on stylometry shows a decreasing trend. Since with increase in the number of users the total classes also increases, this can result in a decrease in the performance of the classifiers to identify the correct class of an instance. When the timeprint features are also incorporated, there is a change in the results computed, both for the Support Vector Machine classifier and the Naive Bayes classifier. There is significant improvement in the detection of the aliases of the users. The top-30 users for combined features give 76% accuracy with the Naive Bayes classifier while for the SVM classifier, they give an accuracy of 92.66% accuracy, as shown in Table III.

When the number of users are increased to 40, the accuracy of the combined case also shows a small decrease to 69.5% for Naive Bayes classifier and 92.5% for the top 50 users, the aliases were identified correctly by Naive Bayes with 66.8% accuracy and with Support Vector Machine by 88.4%. From these results, it can be inferred that the combined technique is more accurate to detect the aliases. A significant increase in the accuracies can be seen when both timeprints and stylometric features are used. As the number of users increases, there is a decrease in the accuracy of both classifiers in detection of the aliases.

From Table 3, it can also be inferred that the performance of the individual stylometric technique for identification of users decreases more in comparison to the combined technique.

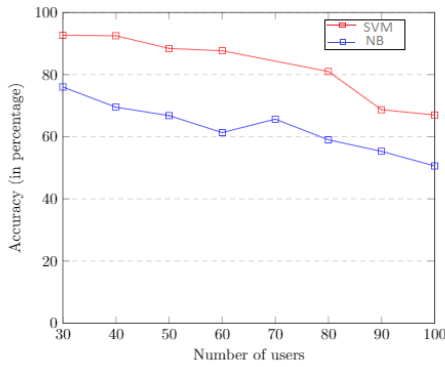


Fig. 2. Accuracy Results for combined technique from classification with NB and SVM classifiers

As for top-100 users, the stylometric technique gives the accuracies of 46.8% by Naive Bayes and 54.4% in the case of the Support Vector Machine classifier but for the combined technique, the accuracies are comparatively much better with 50.6% by Naive Bayes and 67% by the support vector machine classifier. With the increase in the number of users for both techniques, the accuracy shows a decreasing nature but the decrease in the case of the combined scenario is not that steep in comparison to the stylometric technique (refer to Figures 1 and 2).

It can also be observed that the support vector machine classifier outperforms the Naive Bayes classifier. For both stylometry and combined technique, the results given by the Support Vector Machine classifier is much better than the Naive Bayes classifier.

V. CONCLUSION AND FUTURE DIRECTIONS

Multiple aliases of a single user can therefore be identified with better performance by taking into consideration the time profiles of a user. Not only are the write prints crucial but also the timeprints play a vital role as an individual can have a pattern of writing the posts during particular instances of time. From our research we are able to justify that time patterns have a scope for identification of the users over the social networking sites. This analysis can be helpful for cyber security by enabling the identification of the aliases of the extremists in order to put a curb to the usage of sensitive information over the social networking sites.

The authors aim to extend this study for unsupervised learning techniques for identification of the top most likely possible users for which that alias corresponds. Along with the two techniques, the Twitter profiles of the users can also be compared. The inclusion of friends and followers can prove to enhance the performance.

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